

UV-Raman Spectroscopic Identification of Fungal Spores Important for Respiratory Diseases

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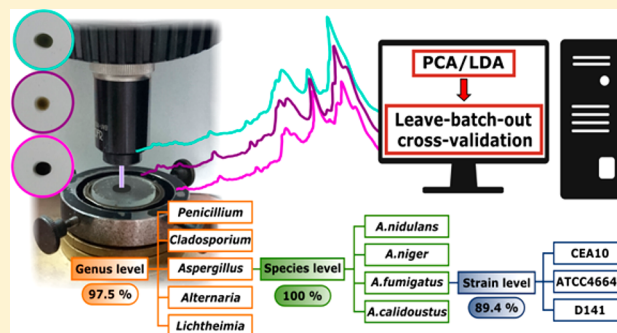
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Supporting Information

ABSTRACT: Fungal spores are one of several environmental factors responsible for causing respiratory diseases like asthma, chronic obstructive pulmonary disease (COPD), and aspergillosis. These spores also are able to trigger exacerbations during chronic forms of disease. Different fungal spores may contain different allergens and mycotoxins, therefore the health hazards are varying between the species. Thus, it is highly important quickly to identify the composition of fungal spores in the air. In this study, UV-Raman spectroscopy with an excitation wavelength of 244 nm was applied to investigate eight different fungal species implicated in respiratory diseases worldwide. Here, we demonstrate that darkly colored spores can be directly examined, and UV-Raman spectroscopy provides the information sufficient for classifying fungal spores. Classification models on the genus, species, and strain levels were built using a combination of principal component analysis and linear discriminant analysis followed by evaluation with leave-one-batch-out-cross-validation. At the genus level an accuracy of 97.5% was achieved, whereas on the species level four different *Aspergillus* species were classified with 100% accuracy. Finally, classifying three strains of *Aspergillus fumigatus* an accuracy of 89.4% was reached. These results demonstrate that UV-Raman spectroscopy in combination with innovative chemometrics allows for fast identification of fungal spores and can be a potential alternative to currently used time-consuming cultivation.



Hundreds of millions of people of all ages suffer every day from chronic respiratory diseases. According to the latest WHO estimates (2004), which are considered conservative, >300 million people are afflicted with asthma, 210 million people suffer from chronic obstructive pulmonary disease (COPD), while >400 million suffer from some other form of respiratory disease.¹ These diseases all have a negative impact on the quality of life of those affected and in many instances can be life threatening, especially in cases of acute exacerbation. Exacerbations of allergic respiratory disease occur frequently in COPD and asthma patients, likely triggered by exposure to environmental stimuli. Although debated, these exacerbations are generally characterized by increased airway inflammation, mucus production, and impaired lung function.

Outdoor fungal spores are one of several environmental factors responsible not only for causing respiratory diseases in humans,^{2–4} but also for triggering exacerbations during chronic forms of lung diseases such as asthma and COPD. For example, it was reported that asthma hospitalization cases increase during thunderstorms due to the increased aerosolization of fungal spores.⁵ Since different fungi may produce different allergens and mycotoxins, the severity of asthma exacerbation may vary between spores of different fungal taxa,⁶ e.g., some patients have hypersensitivity only to *Aspergillus spec.*, which can lead to allergic bronchopulmonary aspergillosis

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(ABPA) or other complications in patients suffering from asthma, cystic fibrosis, or COPD.⁷ Therefore, it would be beneficial to identify and eliminate fungal spores before they trigger respiratory disease or exacerbations in patients with chronic respiratory diseases.

Currently the “gold standard” for fungal identification is the cultivation of sampled organisms coupled with careful observation and measurement of macroscopic and microscopic morphological characteristics of the organism of interest, including reproductive structures like spores. However, this method is laborious and requires highly trained and experienced personnel. Additional difficulties in fungal identification include the presence of a wide range of nonculturable organisms and/or contamination of fungal cultures with fast-growing bacteria, as these factors make the analysis of morphological information complicated.⁸ A fast, highly automated method, which can identify fungal spores via pattern recognition, would exclude the human factor from the identification process and would improve turn-around times for spore identification. As an alternative approach to the more classic culture methods, polymerase chain reaction (PCR)-based detection methods for fungal spore identification have been reported.^{8–11} PCR methods require expensive reagents and design of target-specific primers and tend to produce false-positive results. Another possibility is offered by vibrational spectroscopy, a technique which utilizes molecular vibrations to provide information about the molecular composition, structure and behavior within a sample.

Among the vibrational spectroscopic methods, Raman spectroscopy is a very promising tool for the characterization of microorganisms.^{12,13} This method offers specific molecular information about the chemical species in the samples in a noninvasive and label-free manner. The abilities of Raman spectroscopy for investigation and characterization of airborne allergens, such as individual pollen grains, has been demonstrated by various groups.^{14–16} The characterization of fungal spores using Raman spectroscopy with visible excitation wavelengths has been previously also reported. However, in these studies, the excitation wavelengths utilized could only be applied to white or light-colored spores.^{17,18} Several different microfungi spores relevant to indoor contamination were successfully characterized and identified with Raman spectroscopy, describing the biochemical composition of a single spore.¹⁹ A study of C. Wang et al. reported measurements of Raman spectra in 1600–3400 cm^{-1} spectral range from individual pollen particles and Bermuda grass smut spores held in a photophoretic trap. Due to the small size of spores the spectrum had high background and only three Raman bands were visible.²⁰ Raman spectroscopy with visible excitation wavelengths was also used to discriminate *Aspergillus lentulus* from *Aspergillus fumigatus* with an accuracy of only 78%.¹⁷ K. De Gussem et al. combined Raman spectroscopy with linear discriminant analysis (LDA) and reached around 90% accuracy in assigning the spectra of spores to the correct genus,¹⁸ while identification on the species level was not possible. Instead of applying visible Raman excitation wavelengths various studies on mammalian cells,^{21,22} bacteria,^{23–25} and pollen²⁶ showed that utilizing electronically resonant excitation wavelengths in the UV region can be beneficial for the identification of microorganisms due to a selective resonant enhancement of the Raman signals of taxonomically important macromolecules, e.g., DNA/RNA bases and aromatic amino acids. Furthermore, the fluorescence background in UV resonance Raman spectra

is negligible for excitation wavelengths below 260 nm, which lead to Raman spectra with high signal-to-noise ratio.^{27–31}

In the present study, for the first time the UV-Raman spectroscopic identification of fungal spores is demonstrated. Eight different filamentous fungal species, implicated to a varied extent in respiratory diseases worldwide, were examined: *Aspergillus fumigatus*, *Aspergillus niger*, *Aspergillus nidulans*, *Aspergillus calidoustus*, *Cladosporium herbarum*, *Alternaria alternata*, *Penicillium rubens*, and *Lichtheimia corymbifera*. Except for the mucoralean fungus *Lichtheimia corymbifera*, a basal fungal lineage, all other species belong to the division Ascomycota. In contrast to previous studies,^{17,18} highly pigmented spores were successfully investigated. The identification of spores was based on a combination of principal component analysis (PCA) and LDA of the UV resonance Raman spectral data and was applied at a genus, species, and strain level. The performance of the models was evaluated with leave-one batch-out cross-validation. The presented results highlight the possibility of UV-Raman spectroscopy as a promising method for the automated identification of fungal spores.

MATERIALS AND METHODS

Fungal Spores. For this study 11 fungal strains from 8 different species were utilized, namely *A. fumigatus*, *P. rubens*, *A. niger*, *C. herbarum*, *A. alternata*, *A. nidulans*, *A. calidoustus*, and *L. corymbifera* (Table S-1 of the Supporting Information, SI). The *A. fumigatus* pksP mutant has been described previously.^{32,33} The fungi were grown for 7 days, in the dark, on 10 cm agar plates at room temperature ($\sim 22^\circ\text{C}$). Growth was conducted on *Aspergillus* minimal media (AMM; 6.0 g/L NaNO_3 , 0.52 g/L KCl, 1.52 g/L KH_2PO_4 , 0.52 g/L $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 1% (wt/vol) glucose, and 1 mL of trace element solution [1 g/L $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, 8.8 g/L $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, 0.4 g/L $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 0.15 g/L $\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$, 0.1 g/L $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$, 0.05 g/L $(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}$] per liter),³⁴ malt agar (MA) (Sigma-Aldrich), or modified SUP agar plates (4.1 g/L KH_2PO_4 , 1.1 g/L NH_4Cl , 0.9 g/L K_2HPO_4 , 0.1 g/L MgSO_4 , 1% (wt/vol) glucose, and 0.5% (wt/vol) yeast extract).³⁵ AMM was supplemented with 0.06 mg/L biotin and 5 mM arginine for growth of *A. nidulans* A89. Conidia (asexually produced spores) from each strain were collected in 10 mL of water and separated from hyphae through a 40 μm pore filter. After centrifugation for 10 min at $1800 \times g$, conidia were resuspended in 3% (v/v) formaldehyde and incubated for 1 h for inactivation. Following formaldehyde-inactivation, each conidial-mixture was washed with water to remove any contaminating formaldehyde. The inactivation of the spores was confirmed by the lack of growth on appropriate media. Each growth condition was tested in triplicate, on three separate days. For the UV-Raman measurements 10 μL of spore suspensions was spread on a fused-silica surface and allowed to dry at room temperature.

Spectroscopic Instrumentation. Raman measurements were performed using a LabRam HR 800 spectrometer from Horiba Jobin-Yvon (Bensheim, Germany) with a 2400 lines/mm grating. As excitation wavelength, a frequency doubled argon-ion laser (Coherent Innova 300, MotoFRd, Coherent, Dieburg, Germany) operating at 244 nm was applied. The samples were illuminated via a microscope (Olympus BX 41) equipped with a 20 $\times$ magnification antireflection coated UV objective (LMU UVB) with a numerical aperture of 0.4. The laser power at the laser head was ca. 10.5 mW and on the

sample ca. 0.8 mW at a spot diameter of $\sim 1\ \mu\text{m}$, corresponding to $\sim 10^5\ \text{W}/\text{cm}^2$ irradiance. The entrance slit was set to 300 μm . The Raman scattered light was detected by a nitrogen-cooled CCD camera with an integration time of 20 s with 2 accumulations. The samples were rotated with a speed 60 rotations/min and moved in the x,y direction after each rotation to obtain an average spectrum over a large sample area to minimize possible photodegradation by UV radiation. Ten spectra were collected per sample. Measurements were performed by two independent operators.

Data Analysis. Data processing was performed using an in-house developed script in the programming language R.³⁶ First, the spectra were wavenumber calibrated by using the Teflon Raman spectrum measured prior to each sample measurement as reference. Next, the Raman spectra were background corrected using the sensitive nonlinear iterative peak (SNIP) clipping algorithm³⁷ with a second-order clipping filter. Finally, all spectra were vector normalized and used as input for PCA, which was performed to reduce the dimensionality of the data while retaining the most significant information for classification. PCA was followed by LDA. The performance of the created LDA model for classification of fungal spores was estimated using the leave-one batch-out-cross-validation (LBOCV) approach.³⁸ In this method, one batch was held out from the data set, and the LDA model was redeveloped using the remaining spectra. The resultant model was then used to classify the removed batch. This process was repeated with every batch until all spectra were classified. In this manner LBOCV of the PCA-LDA model was utilized for all number of principle components (PCs). Then, an optimal number of PCs was chosen by finding a saturation point of the accuracy as a function of the number of PCs. For spectral comparison across the groups, mean Raman spectra for each fungal species were calculated using preprocessed, vector-normalized spectra of all batches.

RESULTS AND DISCUSSION

The primary goal of this study was to determine the feasibility of UV-Raman spectroscopy for the identification of melanised fungal spores. The selected fungal spores were all highly pigmented: black for *A. alternata* and *A. niger*, dark green for *A. fumigatus* strains and *P. rubens*, brown for *A. calidoustus* and *C. herbarum* and light gray for *L. corymbifera* and *A. nidulans* (Figure S-1). The difficulties of recording Raman spectra using visible excitation wavelengths from such dark conidia were previously reported; the dark pigments result in a strong interfering fluorescent signal, which masks the original spectrum.¹⁷ Therefore, initially, the possibility of obtaining good quality spectra with UV-Raman from deeply colored, highly absorbing samples was investigated. For this purpose, wild-type *A. fumigatus* conidia were compared to non-pigmented conidia of a strain with a mutation in the *pksP* gene, essential for the formation of the gray-green spores containing dihydroxynaphthalene (DHN)-melanin.³² The UV-Raman spectra of these samples are depicted in Figure 1.

During the measurements, the samples were continuously rotated; however, this rotation was insufficient to completely eliminate the laser-induced degradation of spore proteins into polymeric hydrogenated amorphous carbon. Thus, a carbon background manifesting itself by the two broad bands at approximately 1360 and 1610 cm^{-1} is observed.^{39–41} For the nonpigmented *pksP* mutant strain, the graphitization level was lower, and therefore the bands originating from the spores in

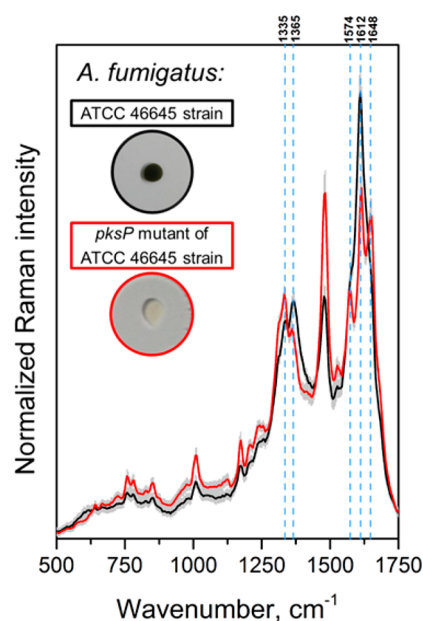


Figure 1. Mean UV-Raman spectra and their double standard deviation of the pigmented *A. fumigatus* ATCC 46645 strain and the nonpigmented *pksP* mutant strain.

the region of 1500–1700 cm^{-1} are more prominent. However, the general shape of the spectra and the peak positions are the same for both samples. In the pigmented wild-type spores of *A. fumigatus*, the unique spore-related spectral features necessary for the spore identification process are still easily visible. Hence, it can be concluded that UV-Raman can be used for the measurement of highly pigmented spores.

To compare a diverse range of allergenic fungi, we had to cultivate these organisms on several different media to promote sporulation due to different nutrient requirements. To test the variability induced by different culture media, we cultivated *P. rubens* strain ATCC 28089 and *A. fumigatus* strain ATCC 46645 on both MA and AMM medium for direct comparison. The mean UV-Raman spectra of the collected spores are presented in Figure S-2. The influence of the culture medium on the spectra was visible and can be best seen in the 1250–1700 cm^{-1} wavenumber region. In the case of *A. fumigatus* grown on MA medium, the signal at 1365 cm^{-1} is more intense, whereas for *P. rubens* the peak at 1648 cm^{-1} is more prominent. Despite this observation, all strains grown on different agars were included into the classification model, proving that the differences from cultivation conditions are smaller than the differences due to species affiliation.

The representative preprocessed mean Raman spectra of the studied fungal spores are depicted in Figure 2.

For all species, primary bands were observed in the wavenumber region between 750 and 1700 cm^{-1} , which is typically associated with various nucleic acids and protein subunits in UV-Raman spectroscopy. The band positions are in good agreement with previously published UV-Raman spectroscopic studies of microorganisms.^{24,23,42} The signal at 1648 cm^{-1} can be assigned to thymine and the one at 1612 cm^{-1} to the aromatic amino acids tyrosine and tryptophan. Guanine and adenine exhibit peaks at 1574 and 1479 cm^{-1} ; they also contribute to the signal at 1335 cm^{-1} . The latter band (at 1333 cm^{-1}) also contains information from tyrosine. The signal at 1526 cm^{-1} can be assigned to cytosine. Thymine and

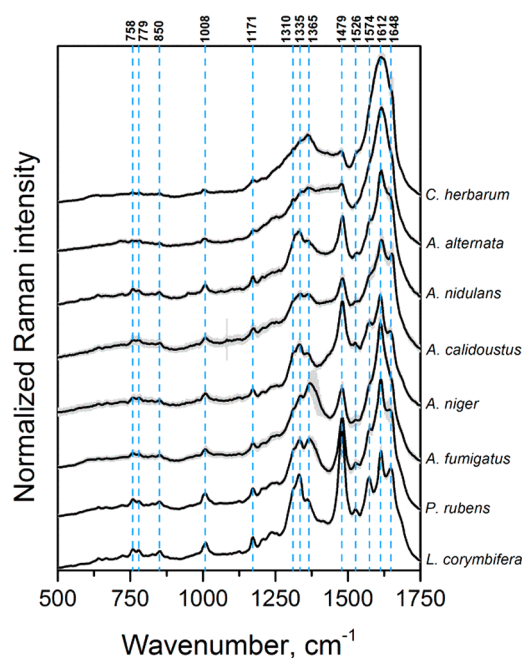


Figure 2. Mean UV-Raman spectra of the all batches from different fungal spores plotted together with double standard deviation (gray area). For the *A. fumigatus* the mean spectrum represents all investigated strains.

adenine exhibit a band at 1365 cm^{-1} . The peak at 1171 cm^{-1} can be assigned to tyrosine. Tryptophan exhibits characteristic bands at 758 and 1008 cm^{-1} . In this study, measurements were performed on the bulk sample under continuous rotation; so several spores contribute to one spectrum. Therefore, one spectrum already comprises metabolic and developmental diversity within the spore population. The reproducibility of these spectra was tested by measuring the Raman spectra from three independent batches.

It is evident from Figure 2 that the fungal spores belonging to different genera display distinct Raman spectral signatures and are easily distinguishable based on their spectra. The peak positions for all analyzed spores are the same, however the relative intensities for the bands in the region of 1250 and 1750 cm^{-1} differ. For the *C. herbarum* and *A. alternata* measurements, three Raman modes at 1574 , 1612 , and 1648 cm^{-1} are strongly convoluted due to a higher carbonization

background. In addition to a visual comparison, a PCA/LDA model was applied and verified using the leave-one batch-out-cross-validation. Three different models on genus, species, and strain level were established (see Figure 3). Health hazards related to fungal spores may differ across genera and species due to the difference between allergenicity and types of mycotoxins produced. It is known for example that the severity of asthma exacerbation may vary between spores of different taxa.⁶ To prevent exacerbations, it is important to define the composition of spores in the air on the genus level. Thus, the classification model was first trained to distinguish spores on the genus level. Prior to LDA, the data size was reduced by PCA. The optimal number of PCs was chosen by finding a saturation point of the accuracy as a function of the number of PCs; in this case, 6 PCs were selected (Figure S-3). The LDA model was trained with 2 batches of spores and then tested with one independent batch of the same strain. This allowed for three different batch permutations for validation and gave a reliable unbiased classification model. The sensitivity, specificity, and accuracy of the LDA model in each run were calculated and averaged. Table 1 shows the number of spectra that were classified correctly for each species of the fungi and also summarizes sensitivity and specificity.

Out of 1079 spectra, 27 were misclassified; this resulted in 97.5% accuracy. In Figure 4, the LDA score plot of the classification model is depicted, whereby each dot represents one spectrum and the ellipsoids correspond to confidence regions for the scores on a level of 95% of the spectral data for each class. The biggest amount of errors appeared to be due to the misclassification of *Aspergillus* species to the closely related genus *Penicillium*. The LDA score plot also nicely reflects the phylogenetic relationship between the different genera: *A. alternata* and *C. herbarum* belong to the same class of Dothideomycetes, while the closely related genera *Penicillium* and *Aspergillus* are grouped into the class Eurotiomycetes. The species *L. corymbifera* belongs to a lower group of fungi, the phylum Zygomycota that diverged early in the evolution of true fungi.⁴³

In the second step, the model distinguishes between four different *Aspergillus* species. Several *Aspergillus* species are able to cause infections like invasive pulmonary aspergillosis or allergic reactions like ABPA, while others are nearly non-pathogenic.⁴⁴ In addition, antifungal drug susceptibility is wide-ranging among even quite similar organisms.⁴⁵ Thus, the

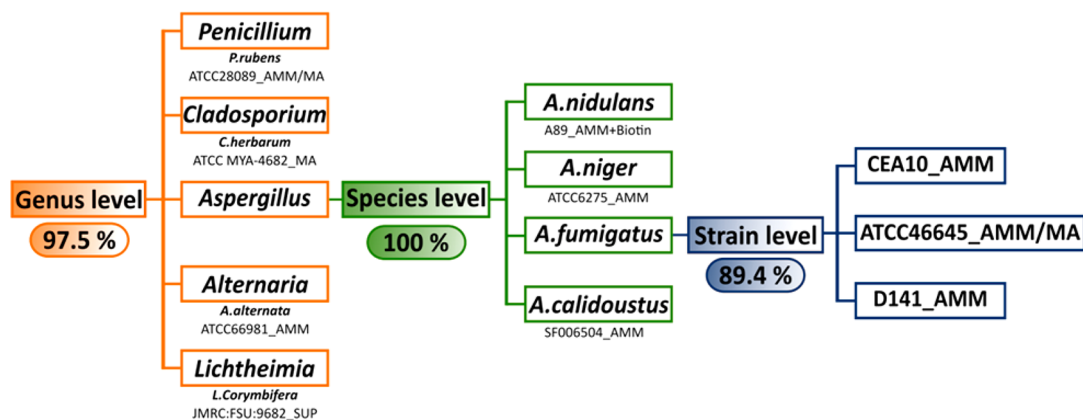


Figure 3. 3-Level PCA/LDA classification model of fungal spores cultured in three independent batches (the percentage represents the accuracies of the model achieved by LBOCV).

Table 1. Identification Results for Different Fungal Spores on the Genus Level

True Predicted	<i>Alternaria alternata</i>	<i>Aspergillus</i> species	<i>Cladosporium herbarum</i>	<i>Lichtheimia corymbifera</i>	<i>Penicillium rubens</i>	Sensitivity, %	Specificity, %
<i>Alternaria alternata</i>	90	0	0	0	0	100	100
<i>Aspergillus</i> species	0	603	0	0	1	95.87	99.78
<i>Cladosporium herbarum</i>	0	4	90	0	0	100	99.6
<i>Lichtheimia corymbifera</i>	0	0	0	90	0	100	100
<i>Penicillium rubens</i>	0	22	0	0	179	99.44	97.55

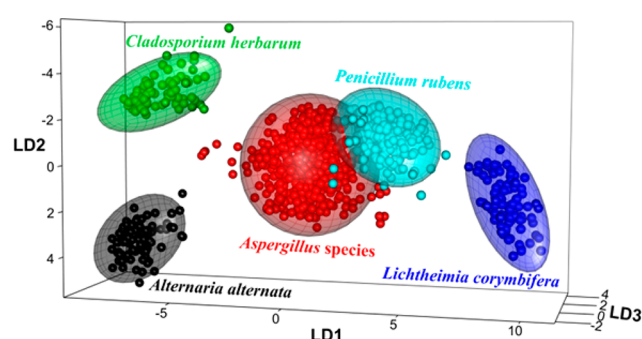
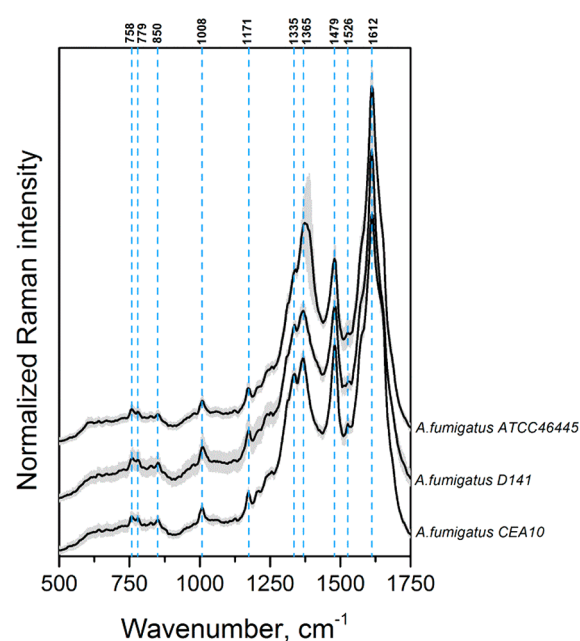


Figure 4. LDA score plots for the classification model on the genus level.

knowledge of the species identity may influence the choice of appropriate antifungal therapy. The visual differentiation of the spectra from *A. fumigatus*, *A. nidulans*, *A. calidoustus*, and *A. niger* is already not as obvious as it was for different genera (Figure 2). For building the identification model on the species level, the number of PCs was set to 7 (Figure S-4). All 629 spectra were assigned correctly resulting in 100% accuracy. The differentiation was based mostly on the different amount of DNA in the spores, mainly resulting in the intensity differences for the peaks at 1365, 1479, and 1648 cm^{-1} (Figure S-5). The achieved accuracy was higher than the one reported by P.E.B. Verwer et al., where the discrimination of *Aspergillus lentulus* from *Aspergillus fumigatus* with VIS-Raman spectroscopy was correct for 78% of spectra.¹⁷

Finally, the model was trained to classify three different strains of *A. fumigatus*, since diversity in virulence among different *A. fumigatus* strains is also well-documented.⁴⁶ The UV-Raman spectra of the three investigated *A. fumigatus* strains are presented in Figure 5. By comparing the spectra, it is clear that the main spectral characteristics of the investigated strains were the same, as a consequence of their close phylogenetic relationship. The LDA model was built with 10 PCs (Figure S-6), and the classification was correct for 89.4% of spectra. The results of this classification model are summarized in Table 2. The misclassification occurred mostly between the ATCC 46445 and D141 strains. The identification of fungal spores on the strain level using Raman spectroscopy was not previously reported.

Figure 5. Mean UV-Raman spectra of three *A. fumigatus* strains.Table 2. Results of the Identification of *A. fumigatus* Strains

True Predicted	ATCC 46445	CA10	D141	Sensitivity, %	Specificity, %
ATCC 46445	151	1	8	83.89	94.97
CEA10	6	89	0	98.89	97.77
D141	23	0	81	91.01	91.48

CONCLUSIONS

In the case of respiratory fungal infections, identification of the fungal agent is often incredibly challenging due to a diverse array of potential organisms and relatively poor diagnostic possibilities. Thus, the development of sensitive and automated methods for the identification of fungal spores is of particular medical interest. In this study, a combination of UV resonance

Raman spectroscopy and chemometrical methods allowed for a highly accurate identification of fungal spores on a genus, species, and even strain level. In all analyzed classification models, strains grown on different agars were included. The promising results presented here indicate that the differentiation of spectra is not determined by the cultivation conditions, but instead based on the differences in the amounts of DNA, RNA, and protein aromatic amino acid units in spores of each species. Moreover, it has been demonstrated that darkly colored spores can be directly examined with UV-Raman spectroscopy, and the contribution of amorphous carbon in the spectra does not inhibit the correct identification of the spores. The obtained classification ratios in the range between 89% and 100% demonstrate that UV resonance Raman spectroscopy represents a powerful tool for fungal spore identification and should be considered for further investigations.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.analchem.8b01038.

Table S-1: Strains used in the study; Figure S-1: Images of dried spore biomass from the studied fungal species on the fused silica substrate; Figure S-2: Mean UV-Raman spectra and their double standard deviation of the *A. fumigatus* (A) and *P. rubens* (B) strains cultured on different media; Figure S-3: Accuracy as a function of the number of PCs for the genus level; Figure S-4: Accuracy as a function of the number of PCs for the species level; Figure S-5: Loading vectors for the species level model; and Figure S-6: Accuracy as a function of the number of PCs for the strain level (PDF)

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The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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